

Bruce Davis

AI Systems Engineer · Agentic AI · IAM · Security

Target roles: AI Systems Engineer · Forward Deployed AI Engineer · Applied AI Engineer · Senior AI Engineer (Agentic Systems) · AI Transformation Lead · Microsoft Copilot Engineer

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SUMMARY

AI Systems Engineer building agentic AI in production environments — MCP servers, ReAct agent loops, multi-agent orchestration, fine-tuned classifiers, Microsoft Copilot deployments, RAG pipelines. Fifteen years of identity and security engineering before that: **Non-Human Identity (NHI) governance at Microsoft**, HIPAA-compliant IAM for 16,000+ users at Veradigm, payment infrastructure for SWIFT/CHIPS/Fedwire at Wells Fargo. Deploying an AI agent safely is mostly an identity and authorization problem. PMP-certified.

SELECTED PROJECTS

Autonomous Security Triage Agent

ReAct agent loop with human-in-the-loop authorization on destructive actions

ReAct agent on Gemini 2.5 Flash — reads logs, enriches threat intel, proposes containment through function-calling tools. Destructive actions pause for explicit human approval. An autonomy dial scopes operational reach. Built for security teams that need AI assistance without surrendering control.

Python · Streamlit · Gemini 2.5 Flash · google-genai SDK · ReAct · Function Calling · HITL.

AI Agent Governance Console

Non-Human Identity governance applied directly to a fleet of AI agents

Six AI agents as discoverable, classified, owned machine identities — scoped credentials, intake-and-approval workflow, rotation, decommissioning in one trail. Generative UI on a posture event stream. The agent fleet itself becomes an NHI governance surface.

Python · React · TypeScript · Multi-agent Architecture · NHI · OAuth · Generative UI.

NHI Credential Risk Classifier

Fine-tuned DistilBERT classifier — 99.33% accuracy, 10.8ms per record on CPU

Synthetic dataset, two fine-tuned models (DistilBERT for production triage, Qwen 2.5 1.5B + LoRA for cachet), full eval harness with per-class P/R/F1, confusion matrix, latency, and cost. Becomes the brain of the Governance Console's Triage agent.

Hugging Face Transformers · PEFT / LoRA · PyTorch · scikit-learn · FastAPI · Python.

Entra ID NHI Audit

PowerShell module that audits an Entra ID tenant for non-human identity risk

Four cmdlets (Connect, Get-Inventory, Test-Risk, Export-Report) score every service principal, app registration, and managed identity on eight risk factors. Mock fixtures so it runs offline; live mode swaps in real Microsoft Graph via managed identity. Self-contained HTML report plus structured JSON for SIEM ingest.

PowerShell 7 · Microsoft Graph SDK · Entra ID · Active Directory · Bicep-ready.

IT Support Copilot for Teams

Teams-deployed Microsoft Copilot — LLM understands, deterministic policy decides

Bot Framework v4 on the Teams side, Azure OpenAI for reasoning, Microsoft Graph for identity. Four tools and four outcomes (self-serve, auto-fulfill, escalate, deny). LLM picks tools; a separate rule-based policy engine decides what happens — auditable and prompt-injection-resistant. Container Apps plus Bicep for production.

Bot Framework · Microsoft Graph · Azure OpenAI · Entra ID · FastAPI · Bicep · Python.

Workflow-Driven Triage Agent

Three protocols, one stack — Google ADK + MCP + n8n

n8n receives a support-ticket webhook, calls a Google ADK agent (Gemini 2.5 Flash) over HTTP; agent calls three MCP tools (intent, customer history, KB) over stdio; n8n routes by priority. Each protocol at the layer it's best for. Docker-compose stack; no-n8n demo path for fast iteration.

Google ADK · MCP / FastMCP · n8n · Gemini 2.5 · FastAPI · Docker Compose.

Linear MCP Server

Production MCP server with documented threat model and 30-case eval harness

Six opinionated MCP tools for engineering team operations — inbox triage, issue scoping, velocity audit, status reporting, 30-case eval harness, documented threat model, per-tool cost analysis. Read-only by design — destructive operations never written. Open source.

TypeScript · @modelcontextprotocol/sdk · Linear GraphQL API · Zod · Vitest.

EXPERIENCE

Business Execution Consultant — Solution Design

Nov 2024 – Apr 2025

Wells Fargo · Contract

- Mapped network dependencies and ran gap analysis across SWIFT, CHIPS, and Fedwire payment infrastructure — documented system flows that hadn't existed on paper, accelerated regulatory approval, and cut design rework by 25%.
- Designed NIST-compliant documentation architecture for network security controls; became the cross-team reference layer for incident response.
- Decomposed executive payment-rail objectives into detailed execution plans for ISO 20022 interoperability and Federal Reserve compliance.

Cloud Solutions Architect — AI

May – Nov 2024

Microsoft · Contract

Forward-deployed Cloud Solutions Architect aligned to Microsoft's AI Factory — embedded with enterprise customers to design, ship, and deploy Azure AI solutions.

- Built Python automation on the Microsoft Graph API for Entra ID Governance — extracted and normalized identity data into executive-ready dashboards and architecture documentation.
- Architected and shipped a Generative AI workflow MVP on Microsoft Copilot — accelerated knowledge transfer and standardized requirements gathering across global delivery teams.
- Translated Non-Human Identity (NHI) governance capabilities into customer-facing solution designs.

Senior Security Engineer Starbucks · Contract	Apr – Sept 2023
- Engineered PowerShell automation across 60+ domains for identity lifecycle and access control remediation — shipped scripts and runbooks at the engineering layer. - Closed 50+ vulnerabilities per sprint through automation and scripted remediation rather than manual triage. - Hardened identity and access controls across distributed teams — configured authentication protocols, owned security testing, and ran defect remediation end to end.	
Project Manager — Privileged Access Management (PAM) Bank of America · Contract	Jul 2021 – Nov 2022
- Led zero-downtime IAM and infrastructure modernization for 12+ teams and 60+ engineers across Oracle, CyberArk, Azure, Middleware, and API platforms — coordinated dependency mapping, change validation, and stakeholder alignment. - Directed HashiCorp Vault enterprise rollout and service account password rotation automation — integration planning, stakeholder communication, and change management. - Owned ServiceNow dashboards as operational source of truth for change approvals, release readiness, and migration tracking; ensured all initiatives met stringent audit requirements. - Authored runbooks, migration plans, and governance documentation for senior leadership; identified delivery risks through deep-dive requirements workshops.	
Senior IAM Security Engineer Veradigm · Healthcare	Jan 2019 – Jun 2021
- Architected and shipped an IAM platform for 16,000+ users in a HIPAA-regulated healthcare environment — Entra ID, MIM, and CyberArk integrated end to end. - Built Power Platform dashboards that surfaced compliance state across the organization; moved audit compliance from 65% to 95% over the engagement. - Designed and executed a multi-year IAM strategy in a regulated healthcare environment, overseeing technology selection and solution architecture.	
Earlier: Data Quality Analyst · Wells Fargo (2017–2018) · IAM Consultant · Cognizant (2016–2017) · Support Escalation Engineer · Microsoft (2012–2015)	

SKILLS

Foundation & Small Models: OpenAI · Anthropic Claude · Google Gemini · Meta Llama · Qwen 2.5 · Mistral · DistilBERT · Hugging Face · Open-source models

ML & Fine-Tuning: PyTorch · Hugging Face Transformers · PEFT / LoRA · scikit-learn · Eval Harness Design · Synthetic Dataset Generation · Per-Class Metrics & Confusion Matrices

Orchestration & Frameworks: MCP · FastMCP · Google ADK · n8n · Microsoft Bot Framework · LangChain · LlamaIndex · Vertex AI · Azure OpenAI · ReAct · Function Calling · Multi-Agent Architecture · RAG · HITL · Anthropic SDK · OpenAI SDK · google-genai SDK · Vercel AI SDK

Application & UI: Microsoft Copilot · Microsoft Teams · Claude · ChatGPT · Gemini · Notion AI

Cloud & Infrastructure: Microsoft Azure · Azure OpenAI · Azure Container Apps · Azure Bot Service · Application Insights · Bicep · Google Cloud Platform · Vercel · Docker · Docker Compose

Identity & Security: Non-Human Identity (NHI) · Entra ID Governance · Microsoft Graph SDK · CyberArk · Microsoft Identity Manager (MIM) · HashiCorp Vault · Active Directory · Zero Trust · HIPAA · NIST · ISO 20022 · SWIFT · CHIPS ·

Fedwire · Threat Modeling · ServiceNow					
Engineering: Python · TypeScript · JavaScript · Node.js · PowerShell 7 · React · Next.js · FastAPI · Streamlit · Pydantic · JSON · JSON Schema · SQL · GraphQL · Zod · Vitest · pytest · Git · Tailwind					
Delivery: PMP · Scrum (PSM I) · Microsoft AI Factory · Forward Deployed Engineering · Audit-Ready Documentation					
CERTIFICATIONS & EDUCATION					
Project Management Professional (PMP) Project Management Institute		M.S., Computer Graphics Technology Purdue University			
Professional Scrum Master I (PSM I) Scrum.org		B.S., Computer Science Lincoln University			
Certified in Cybersecurity (CC) ISC2		Information Systems Carnegie Mellon University			
Google UX Design Professional Certificate Google (via Coursera) · 2023					